Supplementary material: Density-Dependent Vacuum Bubble Nucleation in Matter-Coupled Dark-Sector Cosmology: Dark-Matter Granularity, Wall Dynamics and Transmission

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S1. Methods

granularity.py evaluates the scales of Section 3 (the field mass at the held vacuum, the Compton length, the dark-matter density, the particle, classicality and coherence bounds), the zero-density rate and past-light-cone integral seen between particles, the structural ratio over the bounce tables and toward η_c (with new bounces computed by vacuum_wall.bounce_profile), the protection of the past in regions of reduced density, and the void-by-void completion delay. vacuum_phase_map.py maps the stationary points over density and tilt and evaluates the coherence of the symmetron release. vacuum_wall.py solves the O(4) bounce with its profile, the thin-wall tension with the energy difference removed, the static wall at the degeneracy density, and the nucleation history with the actual Hubble rate. interface_transport.py integrates the mode equation across the planar wall with complex amplitudes (DOP853, rtol 10^{-10}) and computes the reflected momentum flux of a Bose bath with the transverse integral in closed form. bubble_gravity.py evaluates the Israel junction and evolves the enclosed regions by root finding in $\ln a$; bubble_to_cycle.py runs the controls and the viability map; bubble_entropy.py the energy and entropy closure; bubble_handoff.py the percolation integral, the bounds of Table 3, the isolated-bubble test, the collision-product variants and the independent integration; viability_mc.py the survey; interface_validation.py the checks of Table S1. granular_field.py solves the static field in a cell by collocation (solve_bvp, tolerance 10^{-7}) and evaluates the first-order granular rate from the zero-density bounce at full resolution; wave_wall_1d.py integrates the two coupled fields with a leapfrog scheme ($\Delta t = \Delta z / 2$, reflecting ends) at two resolutions; region_sensitivity.py repeats the region evolution and the trapping test with the sudden-crossing energy and other exterior cosmologies; viability_extended.py runs the extended survey and the prediction; env_evolution.py repeats the prediction with evolving contrasts; nonpert_bounce.py finds the bounce with a source on its axis by constrained minimization (L-BFGS, quadratic penalty on the four-volume beyond the barrier top, continuation in the constraint) and solves the static field of one source on a fine radial grid; screened_rate.py recomputes the granular rate with self-screened couplings; neutrino_coupling.py evaluates the optional neutrino coupling and its early-universe history.

Table S1. Validation (interface_validation.py).

Check	Result
degeneracy density theta(deg) = eta/2 (eta = 0.20)	pass
degeneracy density theta(deg) = eta/2 (eta = 0.30)	pass
degeneracy density theta(deg) = eta/2 (eta = 0.36)	pass
degeneracy density theta(deg) = eta/2 (eta = 0.45)	pass
metastable window edges are roots of the cubic discriminant	pass
symmetron: the occupied minimum is never metastable (no tunneling)	pass
coupling check: exact / logarithmic = (exp(d) - 1)/d	pass
kink tension 2/3	pass

kink thickness 4 artanh(0.8)	pass
bounce action at theta = 0 reproduces density bounce (eta 0.20)	pass
bounce action at theta = 0 reproduces density bounce (eta 0.30)	pass
bounce action at theta = 0 reproduces density bounce (eta 0.36)	pass
thin-wall limit: b / B(tw) within 1 % for theta >= 0.6 theta(deg) (eta 0.20)	pass
thin-wall limit: b / B(tw) within 1 % for theta >= 0.6 theta(deg) (eta 0.30)	pass
shooting and collocation agree (eta 0.30, 0.6 theta(deg))	pass
transmission reproduces the Poeschl-Teller result	pass
unitarity T + R = 1	pass
sharp mass step: T = 4 k1 k2 / (k1 + k2) squared	pass
transmission converged in the integration tolerance	pass
junction reduces to the flat-space runaway Gamma = R/R0	pass
wall energy equals the enclosed vacuum-energy deficit	pass
interior bookkeeping dE(in) = F(E) dV + W(wall) closes	pass
wall self-gravity negligible (sigma R / 2 Gamma < 1e-50)	pass
B percolated: H squared = 0 at turnaround and bounce (relative)	pass
B percolated: bounce at rho(c)	pass
B percolated: Hubble rate continuous across nucleation	pass
B percolated: self-similar cycles (ln G constant to 1e-3)	pass
B isolated: H squared = 0 at turnaround and bounce (relative)	pass
B isolated: bounce at rho(c)	pass
B isolated: Hubble rate continuous across nucleation	pass
B isolated: self-similar cycles (ln G constant to 1e-3)	pass
B percolated: comoving entropy never decreases	pass
B isolated: comoving entropy never decreases	pass
C* percolated, fully selective, hot bounce: comoving entropy never decreases	pass
C equals B (computed selectivity changes nothing)	pass
C* fails without rethermalization	pass
percolation completes within 0.1 Hubble time	pass
dark-radiation collision products give the thermalized ln G	pass
bounce density and form leave ln G unchanged	pass
independent initial-value integration reproduces the events	pass

Table S2. Pressure of a thermal test-scalar bath on the wall at the trigger relative to the driving pressure: static wall and ultrarelativistic limit $\Delta m^2 T^2 / 24$*. Temperatures in units of* $\sqrt{\lambda} f$*; the CMB at the trigger is 0.039.*

$g / \sqrt{\lambda}$	**T**	**static**	**relativistic limit**
0.50	0.05	1.3×10^{-8}	4.7×10^{-5}
0.50	0.20	6.8×10^{-5}	7.6×10^{-4}
0.50	1.00	3.6×10^{-3}	1.9×10^{-2}
1.00	0.05	1.3×10^{-11}	1.9×10^{-4}
1.00	0.20	4.1×10^{-5}	3.0×10^{-3}
1.00	1.00	1.1×10^{-2}	7.6×10^{-2}
2.00	0.05	4.5×10^{-18}	7.6×10^{-4}
2.00	0.20	3.2×10^{-6}	1.2×10^{-2}
2.00	1.00	3.0×10^{-2}	3.0×10^{-1}

Table S3. Formal continuation variants (not supported by the dark matter of Section 3; Section 7.3): growth per cycle, bounce temperature ($\rho_r = T^4$) and turnaround relative to nucleation.

Variant	$\ln G$	T_b **(MeV)**	a_{max} / a_{nuc}
baseline (LQC, rho_c = 1e44)	2.6831, 2.6831	187.47	1.230, 1.225, 1.225
rho_c = 1e40	2.6831, 2.6831	18.75	1.230, 1.225, 1.225
rho_c = 1e48	2.6831, 2.6831	1874.66	1.230, 1.225, 1.225
quadratic bounce, H^2 = rho/3 (1 - (rho/rho_c)^2)	2.6831, 2.6831	187.47	1.230, 1.225, 1.225
wall energy as dark radiation until the bounce (w = 1/3)	2.6831, 2.6831	187.47	1.230, 1.225, 1.225
wall energy as massive scalar quanta until the bounce (w = 0)	1.6221, 1.6222	187.47	1.297, 1.292, 1.292

Table S4. Approach to η_c: bounce action, barrier, $\lambda = b / B$ and ρ_{dm} / m_φ^4 at the trigger epoch.

η	θ / θ_{deg}	q	b	**barrier**	λ	ρ_{dm} / m_φ^4
0.360	0.0	0.1	108.5	2.0×10^{-3}	0.378	2.53
0.360	0.0	1.0	108.5	2.0×10^{-3}	0.378	1.39
0.360	0.1	0.1	182.7	4.1×10^{-3}	0.637	0.91
0.360	0.1	1.0	182.7	4.1×10^{-3}	0.637	0.50
0.360	0.3	0.1	448.3	1.0×10^{-2}	1.564	0.20
0.360	0.3	1.0	448.3	1.0×10^{-2}	1.564	0.11
0.370	0.0	0.1	71.0	9.2×10^{-4}	0.248	6.82
0.370	0.0	1.0	71.0	9.2×10^{-4}	0.248	3.75
0.370	0.1	0.1	138.9	2.8×10^{-3}	0.484	1.65
0.370	0.1	1.0	138.9	2.8×10^{-3}	0.484	0.91
0.370	0.3	0.1	373.2	8.4×10^{-3}	1.302	0.29
0.370	0.3	1.0	373.2	8.4×10^{-3}	1.302	0.16
0.375	0.0	0.1	52.4	5.0×10^{-4}	0.183	14.32
0.375	0.0	1.0	52.4	5.0×10^{-4}	0.183	7.88
0.375	0.1	0.1	118.6	2.1×10^{-3}	0.414	2.34
0.375	0.1	1.0	118.6	2.1×10^{-3}	0.414	1.29
0.375	0.3	0.1	339.2	7.6×10^{-3}	1.183	0.35
0.375	0.3	1.0	339.2	7.6×10^{-3}	1.183	0.19
0.380	0.0	0.1	32.6	1.7×10^{-4}	0.114	48.16
0.380	0.0	1.0	32.6	1.7×10^{-4}	0.114	26.49
0.380	0.1	0.1	99.1	1.6×10^{-3}	0.346	3.52
0.380	0.1	1.0	99.1	1.6×10^{-3}	0.346	1.94
0.380	0.3	0.1	307.3	6.8×10^{-3}	1.072	0.42
0.380	0.3	1.0	307.3	6.8×10^{-3}	1.072	0.23
0.383	0.0	0.1	18.3	4.2×10^{-5}	0.064	227.59

0.383	0.0	1.0	18.3	4.2×10^{-5}	0.064	125.18
0.383	0.1	0.1	87.7	1.3×10^{-3}	0.306	4.67
0.383	0.1	1.0	87.7	1.3×10^{-3}	0.306	2.57
0.383	0.3	0.1	289.1	6.3×10^{-3}	1.008	0.48
0.383	0.3	1.0	289.1	6.3×10^{-3}	1.008	0.26
0.384	0.0	0.1	12.0	1.4×10^{-5}	0.042	744.69
0.384	0.0	1.0	12.0	1.4×10^{-5}	0.042	409.58
0.384	0.1	0.1	83.9	1.2×10^{-3}	0.293	5.18
0.384	0.1	1.0	83.9	1.2×10^{-3}	0.293	2.85
0.384	0.3	0.1	283.1	6.1×10^{-3}	0.987	0.50
0.384	0.3	1.0	283.1	6.1×10^{-3}	0.987	0.27

Table S5. Regions of reduced density: ln of the expected number of bubbles in the past light cone for a mean coupling $\theta_0 = 0.3\,\theta_{deg}$, and the smallest mean coupling for which the past is clear.

$1+\delta$	**ln expected bubbles**	**past clear**	**minimum mean θ_0/θ_{deg}**
1.0	-294.0	yes	0.097
0.5	-53.5	yes	0.193
0.3	5.8	no	0.322
0.2	31.2	no	0.483
0.1	55.1	no	0.966

Table S6. Static field in a Wigner–Seitz cell around one source (granular_field.py): field at the source and cell average, and the fraction of the cell within 1% of the zero-density and of the mean-field vacuum (zero-density 0.7865, mean-field 0.7947). The last rows vary the radius of the source.

$R_c m_\varphi$	**source radius**	x **at source**	$\langle x \rangle$	**zero-density**	**mean-field**
0.20	0.01	0.7976	0.7946	0.000	1.000
0.65	0.05	0.8235	0.7938	0.000	0.990
1.96	0.15	0.9264	0.7905	0.000	0.062
6.54	0.30	0.9995	0.7869	0.962	0.006
19.62	0.30	1.0000	0.7865	0.998	0.000
65.41	0.30	1.0000	0.7865	1.000	0.000
196.23	0.30	1.0000	0.7865	1.000	0.000
19.62 (source 0.1)	0.10	1.0000	0.7865	1.000	0.000
19.62 (source 0.3)	0.30	1.0000	0.7865	0.998	0.000
19.62 (source 1.0)	1.00	0.9997	0.7866	0.994	0.001

Table S7. Decay rate with randomly placed point sources at the trigger density: $\ln(\Gamma/\Gamma_0)$ and its fraction of the first-order mean-field suppression (-70.11; the full mean-field value is -86.71). First-order $\partial_\theta b$ = 8859.9 against 8880.1 from the numerical bounce.

m_{dm} **(meV)**	**spacing / Compton length**	$\ln(\Gamma/\Gamma_0)$	**fraction of mean field**
1.0×10^{-4}	9.0×10^{-2}	-69.95	1.0×10^{0}
1.0×10^{-3}	1.9×10^{-1}	-69.9	1.0×10^{0}
1.0×10^{-2}	4.2×10^{-1}	-69.45	9.9×10^{-1}

1.0×10^{-1}	9.0×10^{-1}	-65.24	9.3×10^{-1}
1.0×10^{0}	1.9×10^{0}	-41.09	5.9×10^{-1}
1.0×10^{1}	4.2×10^{0}	-12.21	1.7×10^{-1}
1.0×10^{2}	9.0×10^{0}	-2.829	4.0×10^{-2}
1.0×10^{3}	1.9×10^{1}	-0.572	8.2×10^{-3}
1.0×10^{4}	4.2×10^{1}	-0.1029	1.5×10^{-3}
1.0×10^{5}	9.0×10^{1}	-0.01695	2.4×10^{-4}
1.0×10^{6}	1.9×10^{2}	-0.002602	3.7×10^{-5}
1.0×10^{7}	4.2×10^{2}	-0.0003711	5.3×10^{-6}
1.0×10^{8}	9.0×10^{2}	-4.726e-05	6.7×10^{-7}
1.0×10^{9}	1.9×10^{3}	-5.228e-06	7.5×10^{-8}
1.0×10^{10}	4.2×10^{3}	-5.363e-07	7.7×10^{-9}
1.0×10^{11}	9.0×10^{3}	-5.387e-08	7.7×10^{-10}
1.0×10^{12}	1.9×10^{4}	-5.388e-09	7.7×10^{-11}
1.0×10^{13}	4.2×10^{4}	-5.388e-10	7.7×10^{-12}
1.0×10^{14}	9.0×10^{4}	-5.388e-11	7.7×10^{-13}
1.0×10^{15}	1.9×10^{5}	-5.388e-12	7.7×10^{-14}

Table S8. Convergence of the 1+1-dimensional wall through wave dark matter ($m_{dm}=10\,m_{\varphi}$): arrival time of the wall at $z=150$, fitted Lorentz factor at the end of the run and its ratio to the mean-field run, energy deposited per unit length (sudden 4.188), mean dark-matter velocity behind the wall (predicted 0.7612) and shell compression (predicted 17.5), for four grid spacings and two box sizes.

Δz	box	$t(z=150)$	γ	$\gamma/\gamma_{MF}-1$	deposit	v_{dm}	shell
0.0200	200	143.7633	73.5	-0.0894	4.0052	0.7396	15.03
0.0100	200	143.7292	87.0	-0.0793	4.0460	0.7513	15.93
0.0050	200	143.7153	92.1	-0.0766	4.0733	0.7533	15.83
0.0035	200	143.7121	93.4	-0.0760	4.0858	0.7535	15.85
0.0050	260	143.7153	92.1	-0.0766	4.0733	0.7533	15.83

Table S8b. All runs of the 1+1-dimensional wall through wave dark matter (wave_wall_1d.py): final wall position and Lorentz factor, energy deposited per unit length (in units of the initial dark-matter density; adiabatic 2.72, sudden 4.19) and the mean dark-matter velocity behind the wall (predicted 0.761). Lorentz factors from a fit of hyperbolic motion to the wall position; at the coarser resolution the fitted values change by approximately 6% but their ratio to the mean-field run by less than 0.5%, and the wall position by 0.27%.

run	m/m_{φ}	Δz	z_{wall}	γ	deposit	v_{dm}
wave	10	0.005	195.78	92.1	4.07	0.753
wave	3	0.005	195.78	92.1	4.08	0.750
mean field	–	0.005	195.84	99.8	–	–
none	–	0.005	196.15	142.8	–	–
wave	10	0.010	195.26	87.0	4.05	0.751
mean field	–	0.010	195.32	94.5	–	–
wave	10	0.020	194.21	73.5	4.01	0.740
mean field	–	0.020	194.29	80.7	–	–
wave	10	0.004	195.98	93.4	4.09	0.754

mean field	–	0.004	196.04	101.0	–	–
wave	10	0.005	195.78	92.1	4.07	0.753
mean field	–	0.005	195.84	99.8	–	–

Table S9. Extended survey (viability_extended.py): single fractions T1 (past clear), T2 (completion), T3 (recollapse), T4 (mean field), T5 (runaway with the sudden-crossing energy), and the joint fractions.

priors	draws	T1	T2	T3	T4	T5	T1-3	+T4	+T5
broad	3000	0.686	0.577	0.743	0.043	0.392	0.196	0.0130	0.0053
focused	3000	0.853	0.558	1.000	0.203	0.477	0.410	0.1400	0.1183

Table S10. Scale factor at which a region of density $(1+\delta)$ times the mean completes its transition, with the range for a change of the one-loop prefactor by $(B/2\pi)^{\pm 2}$ ($\eta = 0.30$, $q = 1$, $B_0 = 200$; the grid of scale factors has a spacing of 2.3%).

mean θ_0/θ_{deg}	$1+\delta$	a_{tr}	prefactor range	past clear
1	0.1	1.000	1.000–1.000	yes
1	0.2	1.203	1.176–1.231	yes
1	0.3	1.382	1.350–1.414	yes
1	0.5	1.624	1.587–1.662	yes
1	1.0	2.047	2.000–2.094	yes
1	2.0	2.579	2.520–2.639	yes
2	0.1	1.203	1.176–1.231	yes
2	0.2	1.516	1.481–1.551	yes
2	0.3	1.741	1.701–1.782	yes
2	0.5	2.047	2.000–2.094	yes
2	1.0	2.579	2.520–2.639	yes
2	2.0	3.249	3.174–3.325	yes
3	0.1	1.382	1.350–1.414	yes
3	0.2	1.741	1.701–1.782	yes
3	0.3	2.000	1.954–2.047	yes
3	0.5	2.351	2.297–2.406	yes
3	1.0	2.962	2.894–3.031	yes
3	2.0	3.732	3.646–3.819	yes

Table S11. Isolated Hubble-scale bubble: turnaround, trapping, interior denser than the exterior, restoration and bounce (in H_0^{-1}) for the adiabatic and sudden crossings and three exterior cosmologies (region_sensitivity.py).

case	turnaround	trapped	denser	restoration	bounce
adiabatic, observed	1.256	2.478	2.561	2.696	2.949
adiabatic, Omega_Lambda +0.02	1.256	2.478	2.564	2.696	2.949
adiabatic, Omega_Lambda -0.02	1.256	2.478	2.559	2.696	2.949
adiabatic, held coupling energy in the exterior	1.256	2.478	2.561	2.696	2.949
sudden, observed	1.220	2.318	2.412	2.565	2.849

Table S12. Nonperturbative bounce with a source on the bubble axis (nonpert_bounce.py; grid spacing 0.4, zero-density bounce 438.84 against 441.19 from shooting): action, its increase, the first-order estimate, their ratio and the background field at the source.

q	r_s	b	Δb	first order	ratio	$x_{bg}(0)$
0	0.50	438.84	0.00	0.00	–	0.7865
1	0.50	455.20	16.36	18.66	0.8769	0.8201
3	0.50	478.33	39.49	55.98	0.7054	0.8648
10	0.50	518.61	79.77	186.59	0.4275	0.9327
30	0.50	557.60	118.76	559.76	0.2122	0.9777
100	0.50	589.57	150.73	1865.87	0.0808	0.9958
1000	0.50	611.58	172.74	18658.72	0.0093	0.9999
10000	0.50	614.42	175.58	186587.24	0.0009	1.0000
10000	0.25	501.82	62.98	187481.15	0.0003	0.9999
10000	1.00	745.32	306.48	183971.56	0.0017	0.9999

Table S13. Static coupling of a single source of radius r_s (Yukawa tail of the nonlinear solution, Newton minimization at two radial resolutions) against the thin-shell interpolation $q/(1+q/4\pi r_s)$; a case is accepted if both resolutions converge and agree to 1%.

r_s	q	q_{eff}	interpolation	resolution change	accepted
0.01	0.01	0.0091	0.0093	1.1e-04	yes
0.01	0.1	0.0515	0.0557	1.0e-04	yes
0.01	1	0.1003	0.1116	1.0e-04	yes
0.01	10	0.1180	0.1241	9.9e-05	yes
0.01	100	0.1236	0.1255	8.6e-05	yes
0.01	1000	0.1254	0.1256	1.5e-05	yes
0.01	10000	0.1260	0.1257	5.3e-04	yes
0.1	0.01	0.0099	0.0099	6.6e-05	yes
0.1	0.1	0.0916	0.0926	6.5e-05	yes
0.1	1	0.5240	0.5569	6.3e-05	yes
0.1	10	1.0386	1.1164	6.1e-05	yes
0.1	100	1.2302	1.2410	7.6e-05	yes
0.1	1000	1.2914	1.2551	1.2e-04	yes
0.1	10000	1.3108	1.2565	2.2e-04	yes
0.5	0.01	0.0101	0.0100	6.4e-05	yes
0.5	0.1	0.0995	0.0984	6.4e-05	yes
0.5	1	0.8729	0.8627	6.2e-05	yes
0.5	10	4.0061	3.8587	7.7e-05	yes
0.5	100	6.7643	5.9117	9.7e-05	yes
0.5	1000	7.7781	6.2440	1.2e-04	yes
0.5	10000	8.1088	6.2792	1.9e-04	yes

Table S12b. Convergence of the nonperturbative bounce (source radius 0.5): zero-density action (shooting 441.19) and the increase of the action for $q=10$, 10^3 and 10^4 against grid spacing and box size. The saturated increase falls as the source is resolved and is an upper bound.

grid spacing	box	b_0	$\Delta b(q=10)$	$\Delta b(q=10^3)$	$\Delta b(q=10^4)$
0.40	20	438.84	79.77	–	175.58
0.30	20	439.87	74.79	–	127.76
0.25	20	440.27	67.18	–	104.62
0.40	26	438.84	79.77	–	175.56
0.25	20	440.27	–	104.05	104.62
0.20	20	440.60	–	138.61	140.39

Table S18. Three-dimensional bounce with a periodic row of discrete sources ($q = 1000$, radius 1, spacing d) and with the same coupling per unit length smeared along the line (multisource_bounce.py; grid spacing 0.5). Last column: range of the background field on the axis. The single-source axisymmetric code gives $\Delta b = 298.5$ for one source at the bubble center.

sources	d	b	Δb	axis background
none	–	438.33	0.00	0.786–0.786
discrete	8	679.82	241.50	0.791–0.999
smeared	8	987.00	548.67	0.977–0.995
discrete	12	509.23	70.90	0.787–0.999
smeared	12	979.97	541.64	0.967–0.991
discrete	20	441.62	3.29	0.787–0.999
smeared	20	966.98	528.66	0.951–0.984

Table S14. Granular decay rate with self-screened particles (screened_rate.py): coupling, de Broglie radius, screening factor and $|\ln(\Gamma/\Gamma_0)|$ with and without screening.

m_{dm} (meV)	q	r_s	q_{eff}/q	screened	unscreened
2.5×10^{2}	9.9×10^{1}	1.8×10^{1}	6.9×10^{-1}	1.4×10^{0}	1.5×10^{0}
5.0×10^{3}	2.0×10^{3}	8.9×10^{-1}	5.5×10^{-3}	3.6×10^{-2}	1.7×10^{-1}
1.0×10^{5}	4.0×10^{4}	4.4×10^{-2}	1.4×10^{-5}	4.9×10^{-4}	1.7×10^{-2}
2.0×10^{6}	8.1×10^{5}	2.2×10^{-3}	3.4×10^{-8}	2.3×10^{-6}	1.4×10^{-3}
4.1×10^{7}	1.6×10^{7}	1.1×10^{-4}	8.3×10^{-11}	5.8×10^{-9}	1.1×10^{-4}
8.2×10^{8}	3.3×10^{8}	5.4×10^{-6}	2.1×10^{-13}	1.4×10^{-11}	6.3×10^{-6}
1.7×10^{10}	6.6×10^{9}	2.7×10^{-7}	5.1×10^{-16}	3.5×10^{-14}	3.3×10^{-7}
3.3×10^{11}	1.3×10^{11}	1.3×10^{-8}	1.2×10^{-18}	8.7×10^{-17}	1.6×10^{-8}
6.7×10^{12}	2.7×10^{12}	6.6×10^{-10}	3.1×10^{-21}	2.2×10^{-19}	8.0×10^{-10}
1.4×10^{14}	5.4×10^{13}	3.3×10^{-11}	7.6×10^{-24}	5.3×10^{-22}	4.0×10^{-11}

Table S15. Transition order with evolving density contrasts (env_evolution.py).

$1+\delta$ today	a_{tr} frozen	a_{tr} evolving	$1+\delta$ at a_{tr}	shift
0.10	1.382	1.317	0.087	-0.047
0.20	1.741	1.633	0.165	-0.062
0.30	2.000	1.876	0.247	-0.062
0.50	2.351	2.235	0.429	-0.049
1.00	2.962	2.962	1.000	+0.000
2.00	3.732	4.214	2.880	+0.129

Table S16. Early-universe history for the optional neutrino coupling: scalar mass, neutrino temperature (meV) and the ratio of the neutrino–neutrino scattering rate to the expansion rate per y^4 (contact estimate).

z	m_φ (meV)	T_ν (meV)	m_φ/T_ν	$\Gamma/H\,y^4$
1	9.37	0.335	28	4.6×10^{22}
2	13	0.433	30	3.2×10^{22}
3	20	0.589	34	1.8×10^{22}

4	33.2	0.835	40	8.0×10^{21}
6	58.7	1.23	48	3.2×10^{21}
10	108	1.84	59	1.1×10^{21}
16	205	2.82	73	4.0×10^{20}
25	395	4.38	90	1.3×10^{20}
40	772	6.84	113	4.3×10^{19}
63	1.52e+03	10.7	142	1.4×10^{19}
100	3.01e+03	16.9	178	4.5×10^{18}
158	5.97e+03	26.7	223	1.4×10^{18}
251	1.19e+04	42.3	281	4.4×10^{17}
398	2.36e+04	66.9	353	1.4×10^{17}
631	4.71e+04	106	444	4.2×10^{16}
1000	9.38e+04	168	559	1.3×10^{16}
1585	1.87e+05	266	704	3.8×10^{15}
2512	3.73e+05	421	886	1.1×10^{15}
3981	7.44e+05	667	1115	3.1×10^{14}
6310	1.49e+06	1.06e+03	1404	8.7×10^{13}
10000	2.96e+06	1.68e+03	1768	2.3×10^{13}
15849	5.91e+06	2.66e+03	2225	6.2×10^{12}
25119	1.18e+07	4.21e+03	2802	1.6×10^{12}
39811	2.35e+07	6.67e+03	3527	4.1×10^{11}
63096	4.7e+07	1.06e+04	4440	1.0×10^{11}
100000	9.37e+07	1.68e+04	5590	2.7×10^{10}
158489	1.87e+08	2.66e+04	7037	6.7×10^{9}
251189	3.73e+08	4.21e+04	8859	1.7×10^{9}
398107	7.44e+08	6.67e+04	11153	4.3×10^{8}
630957	1.48e+09	1.06e+05	14041	1.1×10^{8}
1000000	2.96e+09	1.68e+05	17676	2.7×10^{7}

Table S17. Physical parameters of the draws that satisfy every condition (viability_extended.py).

priors	parameter	min	10%	median	90%	max
broad	B0	72.7	99.8	240	265	278
broad	local theta_0/theta_deg	0.00204	0.128	0.437	6.37	7.77
broad	q	0.00571	0.487	2.12	2.82	2.96
broad	1+delta_v	0.0524	0.0662	0.211	0.455	0.468
broad	m_dm (meV)	15.7	20	46.2	120	132
broad	m_phi (meV)	1.33	1.52	2.63	4.32	6.75
broad	a at completion (emptiest regions)	1	1.07	2.07	3.66	6.65
focused	B0	150	180	241	278	287

focused	local theta_0/theta_deg	0.0112	0.0372	0.193	1.33	2.87
focused	q	0.101	0.135	0.54	2.08	2.99
focused	1+delta_v	0.1	0.127	0.213	0.287	0.3
focused	m_dm (meV)	6.33	13.5	38.3	107	202
focused	m_phi (meV)	1.11	1.24	1.83	5.3	6.95
focused	a at completion (emptiest regions)	1	1	1.59	3.45	9.19

S2. Reproduction

Environment of the published run: Python 3.11.9, NumPy 2.4.6, SciPy 1.17.1 (Windows). From the code directory: vacuum_phase_map.py, vacuum_wall.py, interface_transport.py (approximately 25 minutes), bubble_gravity.py, bubble_to_cycle.py, bubble_handoff.py, viability_mc.py (approximately 15 minutes), bubble_entropy.py, interface_validation.py, granularity.py, granular_field.py, wave_wall_1d.py (approximately 3 minutes), region_sensitivity.py, viability_extended.py (approximately 5 minutes), env_evolution.py, nonpert_bounce.py (approximately 7 minutes; 'convergence' and 'screening' modes), multisource_bounce.py (approximately 20 minutes), screened_rate.py, neutrino_coupling.py, make_int_figures.py, make_gran_figure.py, make_resolved_figures.py, paper2_build.py. Input tables shared with the companion paper (bounce_tables.json, thermal_tables.json, survival.json, density_bounce.json, coupling.json, continuous_cycles.json, cycle_map.json) are included as data and documented in the README. Code: https://github.com/sandlerleon/matter-coupled-vacuum-bubbles.